EXERCISE 5

General Solution of an Inhomogeneous System

Original statement · probls_ch4-i_2627_v2.pdf

Inhomogeneous System • Particular Solution

STATEMENT

Suppose that we have an inhomogeneous system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)x + b_1(t)y + f_1(t), \\ \dfrac{dy}{dt} = a_2(t)x + b_2(t)y + f_2(t) \end{cases} \tag{4}$$

and one particular solution

$$\begin{cases} x = x_p(t) \\ y = y_p(t). \end{cases} \tag{5}$$

Show that, if $x_1(t)$ and $x_2(t)$ are linearly independent solutions of the homogeneous system, then the general solution of (4) is

$$\begin{cases} x = c_1x_1(t) + c_2x_2(t) + x_p(t) \\ y = c_1y_1(t) + c_2y_2(t) + y_p(t) \end{cases}$$

SETUP AND HYPOTHESES

Coefficients and forcings f_1, f_2 continuous on $[a,b]$. The homogeneous system is (1) of Exercise 2. $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$ are linearly independent solutions of (1) (the statement writes "$x_1(t)$ and $x_2(t)$ " for the two solution vectors). Their Wronskian is non-zero on $[a,b]$: by Exercise 3 it is either $\equiv 0$ or never zero, and if $W(t_0) = 0$ there would be $(c_1, c_2) \neq (0,0)$ with $c_1\mathbf{x}_1(t_0) + c_2\mathbf{x}_2(t_0) = \mathbf{0}$; the combination $c_1\mathbf{x}_1 + c_2\mathbf{x}_2$ solves (1) with zero data at t_0, so it is $\equiv \mathbf{0}$ by uniqueness (Exercise 4, Step 2), contradicting independence. $\mathbf{x}_p = (x_p, y_p)$ solves (4).

FORMULATION

Two claims: (A) every function of the displayed form solves (4); (B) every solution of (4) has that form. The key idea (Robinson Sec. 14.1 for scalar equations, Topic 4 Sec. 3.1) is that the difference of two solutions of (4) solves the **homogeneous** system.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The last theorem of the notes for linear systems: $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$. The proof is the one of the scalar case (Topic 3, notes p. 6): the difference of two solutions of the inhomogeneous system solves the homogeneous one.

DERIVATION

(A) The family solves (4)

Write $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p$ and $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ with $A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$, $\mathbf{f} = (f_1, f_2)^T$. Using $\dot{\mathbf{x}}_j = A\mathbf{x}_j$ and $\dot{\mathbf{x}}_p = A\mathbf{x}_p + \mathbf{f}$:

$$\dot{\mathbf{x}} = c_1 A\mathbf{x}_1 + c_2 A\mathbf{x}_2 + A\mathbf{x}_p + \mathbf{f} = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p) + \mathbf{f} = A\mathbf{x} + \mathbf{f}$$

(B) Every solution has this form

Let $\mathbf{x} = (x, y)$ be any solution of (4) and put $\mathbf{z} = \mathbf{x} - \mathbf{x}_p$. Then

$$\dot{\mathbf{z}} = \dot{\mathbf{x}} - \dot{\mathbf{x}}_p = (A\mathbf{x} + \mathbf{f}) - (A\mathbf{x}_p + \mathbf{f}) = A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{z}$$

so $\mathbf{z}$ solves the homogeneous system (1). Since $\mathbf{x}_1, \mathbf{x}_2$ are independent solutions of (1) (non-zero Wronskian), Exercise 4 gives constants c_1, c_2 with $\mathbf{z} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2$. Hence $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p$. ■

INTERPRETATION AND CHECKS

- The structure "general = complementary + particular" is the same as for scalar linear equations (Topic 3 Sec. 5 lemma). The two constants are fixed by an initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$: $\mathbf{c} = \Phi(t_0)^{-1}\,[\mathbf{x}_0 - \mathbf{x}_p(t_0)]$.
- Example (Topic 4 Example 26.1): $\dot{x} = x + y$, $\dot{y} = 4x - 2y + 4e^{-2t}$ has the particular solution $(-e^{-2t}, 3e^{-2t})$: check $\dot{x}_p = 2e^{-2t} = x_p + y_p = -e^{-2t} + 3e^{-2t}$ and $\dot{y}_p = -6e^{-2t} = 4(-e^{-2t}) - 2(3e^{-2t}) + 4e^{-2t} = -6e^{-2t}$.
- A particular solution can be found by elimination (Exercise 7), undetermined coefficients, or variation of parameters $\mathbf{x}_p = \Phi \int \Phi^{-1}\mathbf{f}$ (Topic 4 Sec. 3.1).
- Different particular solutions differ by a homogeneous solution, which is absorbed in c_1, c_2.